First Term Examination, 2014-15
Course: B.Tech (CSE), Year: II, Semester: IV
Operating Systems (CSE 221)

Uni. Roll No: .....

Time: 1.5 Hrs. **Total Marks: 20**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x5=5)**

I. Define Spooling.

II. What are the System Calls in UNIX and Windows defined for Process Control.

III. What are the different elements of process image?

IV. Define Processor Affinity.

V. Define Context Switching.

Section B

Note: Attempt any Three Questions **(2x3=6)**

I. Explain any four reasons for process creation and process termination in operating system.

II. The services provided by an operating system can be divided into two main categories. Briefly describe the two categories and how they differ.

III. What are the similarities between thread and a process? Differentiate between User-level thread and Kernel-level thread.

IV. Explain Layered and Microkernel structures of the Operating System.

Section C

Note: Attempt any Three Questions **(3x3=9)**

I. What is a process? With the help of process state transition diagram explain process state model with two suspended states in detail. Also draw the queuing diagram.

II. Consider the following set of processes, with the length of the CPU burst given in milliseconds:

Process	Burst Time(ms)	Priority
P1	10	3
P2	1	1
P3	2	3
P4	1	4
P5	5	2

The processes are assumed to have arrived in the order P1, P2, P3, P4, and P5 all at time 0. Draw Gantt charts that illustrate the execution of these processes using the following scheduling algorithms:

(a) Non-preemptive Priority (consider 1 as highest priority and 4 as lowest priority)

(b) Round-Robin (q=2)

What is the average turnaround time and average waiting time for each of the scheduling algorithms?

III. What is PCB? Explain all its components in detail.

IV. Write short note on:

(a) Multi-level Feedback Queue Scheduling.

(b) Difference between Multiprogramming and Multiprocessing.

Printed Pages-2 **University Roll No.....**

B. Tech. II Year IV Semester, First Term Examination, 2014-15

CSE-222: Introduction to Microprocessor

Time: 01 ½ Hours **Total Marks: 20**

Note:

1. Attempt **all** questions from Section-A, **any three** from Section-B and **any three** from Section-C.
2. Symbols have their usual meanings.

Section-A

Note: Attempt all Questions **(1*5=5)**

I. Which microprocessor pins are used to request and acknowledge a DMA transfer?

II. How many Total T-states required for STA instruction? Justify your answer.

III. Discuss Compare instruction with example in 8085 µp.

IV. What is the name of machine cycle if status pin are IO/M =0, S_1=0 and S_0=0?

V. Discuss relationship between Instruction cycle, Machine cycle and T-state with diagram.

Section-B

Note: Attempt any three questions **(2*3=6)**

I. What are the differences between a microprocessor and microcontroller?

II. Explain the need to Multiplex and De-Multiplex of the bus AD_7-AD_0 with neat diagram in 8085 µp.

III. Write an assembly language program to add two 16 bits number available in BC & DE register pair, store the result in HL register pair. Calculate the execution time of the program for 8085 µp.

IV. Write 8085 instructions for
(a) Load 2010H data to HL Register Pair.
(b) Transfer contents of 1050H memory location to B Register.
(C) Add 40H as immediate data with contents of accumulator.
(d) Add contents of accumulator with accumulator.

Section-C

Note: Attempt any three questions **(3*3=9)**

I. What is need of addressing modes in microprocessor? Write and explain addressing modes of 8085 with example.

II. Define Machine cycle? Draw the timing diagram for LXI instruction.

III. Draw architecture of basic microprocessor and explain its all components.

IV. Explain the following instructions with suitable example. Also indicate the flags (CY, S & Z) condition.
(a) ADI (b) XOR (c)LHLD

Printed Pages-2 University Roll No.....

B. Tech. II Year IV Semester, First Term Examination, 2014-15

CSE-223: Computer Graphics

Time: 01 ½ Hours Total Marks: 20

Note:

1. Attempt **all** questions from Section-A, **any three** from Section-B and **any three** from Section-C.
2. Symbols have their usual meanings.

Section-A

Note: Attempt all Questions (1*5=5)

I. An image has a height of 3 inches and an aspect ratio of 2.5. What is its width?

II. Find seven other symmetric points of a point (6, 7) that is lying on a circle whose center is origin.

III. What is horizontal retrace and vertical retrace? Explain.

IV. Consider a raster scan system with resolution of 1280x1024. What size of frame buffer (in bytes) is needed for the system to store 12 bits/pixels?

V. What is persistence? Explain.

Section-B

Note: Attempt any three questions (2*3=6)

I. Enlists at least four differences between Raster scan system and Random scan system.

II. If a polygon have multi colour boundary, then which Algorithm is more suitable to fill the colour? Write & Discuss that algorithm in detail.

III. A Raster scan system has resolution of 640x480. How many pixels could be accessed per second by a display controller that refreshes the screen at a rate of 60 frames per second?

IV. Write Mid point circle drawing algorithm and execute it for radius r=5, centered on (0, 0).

Section-C

Note: Attempt any three questions **(3*3=9)**

I. Explain the working of DVST and compare it with refresh CRT.

II. Discuss convex and concave polygons with suitable example. Write and explain Weiler-Atherton polygon clipping algorithm.

III. Write Bresenham's line drawing algorithm. Consider a line from (5, 5) to (13, 9) by using Bresenham's line drawing algorithm rasterizing the line.

IV. Write Liange Barsky line clipping algorithm and find the visible portion of the line segment A(-1/2,6) B(9,1) while Xmin=1, Xmax=7, Ymin=2 and Ymax=6 given.

Printed Pages-2 **University Roll No.....**

B. Tech. (CSE) II Year IV Semester,

First Term Examination, 2014-15

CSE224: Computer Networks

Time: 01 ½ Hours **Total Marks: 20**

Note:

1. Attempt **all** questions from Section-A, **any three** from Section-B and **any three** from Section-C.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.

Section A

Note: Attempt All Questions **(1x5=5)**

I. Apart from difference in the number of layers, mention at least two differences between OSI and TCP/IP protocol stack.

II. What is the use of modulation? Name different types of digital to analog modulation techniques.

III. In which Medium Access Control method, after the station finds the line idle it sends or refrain from sending based on the outcome of a random number generator. If the line is busy, it tries again.

IV. Represent 101100101011 in Differential Manchester Encoding.

V. Define the function of the following networking devices.
Hub, Switch, Router and Gateway.

Section B

Note: Attempt any Three Questions **(2x3=6)**

I. Explain the working of CSMA/CD medium access control protocol.

II. Differentiate between Circuit switching and Packet switching indicating the total delay possible in both cases.

III. Explain the functionalities of at least six layers of OSI protocol stack.

IV. In a digital transmission, the sender clock is 0.2 percent faster than the receiver clock. How many extra bits per second does the sender send if the data rate is 1 Mbps? What will be the baud rate if the encoding followed are NRZ-I and Differential Manchester?

Section C

Note: Attempt any Three Questions **(3x3=9)**

I.

a. What are the throughput of pure ALOHA and slotted ALOHA?

b. Assume CSMA/CD protocol. Find the minimum frame length for a 1Mbps bit rate and maximum network span of 10 kilometers with no repeaters. Assume a medium propagation delay of 4.5 nanoseconds per meter.

II. A computer monitor has a resolution of 1200 by 1000 pixels. If each pixel uses 10 bits, how many bits are needed to send the complete contents of a screen? How much time will be required to send these bits over a telephone line (4 KHz of bandwidth) and the noise is 5 mV when the signal is 10 V? What will be the signal levels used for the transmission?

III. Explain the working of CDMA with an example involving four nodes. Your answer should also describe how the codes for these nodes can be formulated.

IV. Write a short note on different Physical Media that can be used for networking.

First Term Examination, 2014-15

Course: - B.Tech *II-Year, IV-Sem.*

Database Management System (ITE221)

Uni. Roll No: .....

Time: 1.5 Hrs. **Total Marks: 20**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x5=5)**

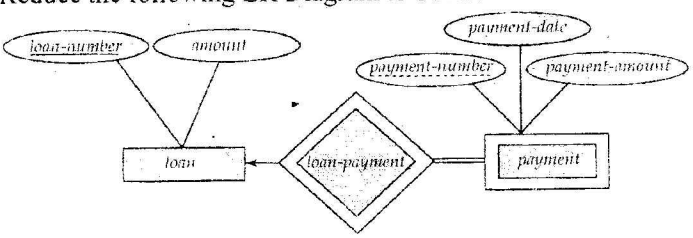
I. What is meant by a safe expression in relational calculus?

II. Write down the constraints that have to follow in Set operations of Relational Algebra?

III. What is difference between database state and database schema?

IV. Why duplicate tuples are not allowed in a relation?

V. Reduce the following ER Diagram to Tables.



Section B

Note: Attempt any Three Questions **(2x3=6)**

I. What do you understand by data independence? Explain the different type of data independence

II. Discuss the entity integrity and referential integrity constraints.

III. What is meant by recursive relationship? Explain it with the help of suitable example.

IV. Draw the ER diagram for a BANK database consider each bank can have multiple branches, and each branch can have multiple accounts and loans.

Section C

Note: Attempt any Three Questions **(3x3=9)**

I. Consider the following schema, and write the following query in Relational Algebra

SUPPLIER (SUPPLIER_ID, SUPPLIER_NAME, SUP_ADD)
PARTS (PART_ID, PART_NAME, COLOUR)
CATALOG (SUPPLIER_ID, PART_ID, COST)

a) Find the name of the suppliers who supply yellow parts.
b) Find the name of suppliers who supply both red and blue parts.
c) Find the name of the suppliers who supply all parts.
d) Find the average cost of Parts.
e) Remove the record of the supplier whose name is John.
f) Add one record in SUPPLIER relation <2001. 'Smith', 'Delhi'>.

II. What is difference between specialization and generalization? And also explain it with example?

III. What are different types of database users? Explain the role of DBA.

IV. Describe the three schema architecture. Why do we need mapping between schema levels?

First Term Examination,
Even Semester 2014-15

Course: B. Tech. II Yr. **Univ. No.** ----------
Title of Paper: Environmental Studies
Paper Code: CEE-201
Time: 1.5 Hours **Max. Marks:** 20

Section-A

Note: Attempt All Questions **1 X 5 = 5 Marks**

(I) What are four segments of environment?
(II) Define synecology with example.
(III) Differentiate lentic and lotic ecosystems
(IV) What do you mean by energy plantation?
(V) What is mineral spoil?

Section-B

Note: Attempt Any three Questions **2 x 3 = 6 Marks**

(I) Write major cause of deforestation. Discuss its consequences.
(II) Discuss the exploitation of mineral resources.
(III) Describe the structure and function of pond ecosystem.
(IV) Discuss water born and water induced diseases.

Section-C

Note: Attempt Any Three Questions **3 x 3 = 9 Marks**

(I) Classify the atmosphere on the basis of temperature variation. States main activities taking place in each of them with diagram.
(II) What are the scopes of environmental science.
(III) Discuss the fluoride problem in drinking water.
(IV) What do you mean by mineral resources? How can you conserve them.